

Chapter 14

Basics of Functional Dependencies and Normalization for Relational Databases



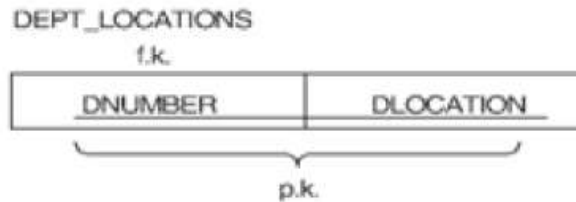
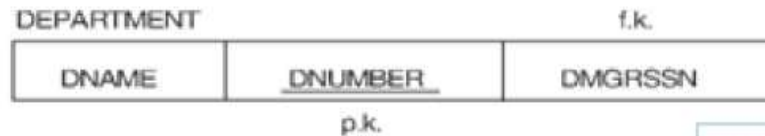
INFORMAL DESIGN GUIDELINES FOR RELATION SCHEMAS

- Four informal guidelines used as measures to determine the quality of relation schema design:
 1. Making sure that the semantics of the attributes is clear in the schema
 2. Reducing the redundant information in tuples
 3. Reducing the NULL values in tuples
 4. Disallowing the possibility of generating spurious tuples

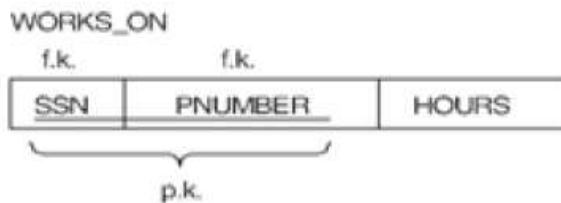
1. Semantics of the attributes

- When we form a relation schema, the attributes belonging to that relation should have certain real-world meaning and a proper interpretation.
- If the conceptual design is done carefully followed by a systematic mapping into relations , the resulting design should have a clear meaning.
- In general, the easier it is to explain the semantics of the relation, the better the relation schema design will be.

- To illustrate this, consider , a simplified version of the COMPANY relational database schema



All the relation schemas here may be considered as easy to explain and hence good from the standpoint of having clear semantics.



We can thus formulate the following informal design guideline.

GUIDELINE 1:

Design a relation schema so that it is easy to explain its meaning. Do not combine attributes from multiple entity types and relationship types into a single relation.

An Example poor relational schema designs

(a)

EMP_DEPT

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER	DNAME	DMGRSSN
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(b)

EMP_PROJ

<u>SSN</u>	<u>PNUMBER</u>	HOURS	ENAME	PNAME	PLOCATION
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- Combining EMPLOYEE and DEPARTMENT into EMP_DEPT
- Grouping attributes from EMPLOYEE, PROJECT and WORK_ON into EMP_PROJ

So the violation of guideline 1

2. Redundant Information in Tuples and Update Anomalies

- One goal of schema design is to minimize the storage space that the base relations (files) occupy.
- Grouping attributes into relation schemas has a significant effect on storage space.

Let us compare the space used by the two base relations EMPLOYEE and DEPARTMENT with that for an EMP_DEPT base relation in the following database states

A Database state for EMPLOYEE

EMPLOYEE				
ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER
Smith,John B.	123456789	1965-01-09	731 Fondren,Houston,TX	5
Wong, Franklin T.	333445555	1955-12-08	638 Voss,Houston,TX	5
Zelaya,Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4
Wallace,Jennifer S.	987654321	1941-06-20	291 Berry,Bellaire, TX	4
Narayan,Remesh K.	666884444	1962-09-15	975 Fire Oak,Humble, TX	5
English,Joyce A.	453453453	1972-07-31	5631 Rice,Houston, TX	5
Jabbar,Ahmad V.	987987987	1969-03-29	980 Dallas,Houston, TX	4
Borg,James E.	888665555	1937-11-10	450 Stone,Houston, TX	1

A Database state for DEPARTMENT

DEPARTMENT		
DNAME	<u>DNUMBER</u>	DMGRSSN
Research	5	333445555
Administration	4	987654321
Headquarters	1	888665555

The same Database state for EMP_DEPT

EMP_DEPT

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER	DNAME	DMGRSSN
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

- In EMP_DEPT, the attribute values pertaining to a particular department (DNUMBER, DNAME, DMGRSSN) are repeated for *every employee who works for that department*.
- In the EMPLOYEE relation only the department number (DNUMBER) is repeated for each employee who works in that department.

A Database state for WORKS_ON and PROJECT

WORKS_ON

<u>SSN</u>	<u>PNUMBER</u>	HOURS
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	null

PROJECT

<u>PNAME</u>	<u>PNUMBER</u>	PLOCATION	DNUM
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

The same Database state for EMP_PROJ

EMP_PROJ					
SSN	PNUMBER	HOURS	ENAME	PNAME	PLOCATION
123456789	1	32.5	Smith,John B.	ProductX	Bellaire
123456789	2	7.5	Smith,John B.	ProductY	Sugarland
666884444	3	40.0	Narayan,Ramesh K.	ProductZ	Houston
453453453	1	20.0	English,Joyce A.	ProductX	Bellaire
453453453	2	20.0	English,Joyce A.	ProductY	Sugarland
333445555	2	10.0	Wong,Franklin T.	ProductY	Sugarland
333445555	3	10.0	Wong,Franklin T.	ProductZ	Houston
333445555	10	10.0	Wong,Franklin T.	Computerization	Stafford
333445555	20	10.0	Wong,Franklin T.	Reorganization	Houston
999887777	30	30.0	Zelaya,Alicia J.	Newbenefits	Stafford
999887777	10	10.0	Zelaya,Alicia J.	Computerization	Stafford
987987987	10	35.0	Jabbar,Ahmad V.	Computerization	Stafford
987987987	30	5.0	Jabbar,Ahmad V.	Newbenefits	Stafford
987654321	30	20.0	Wallace,Jennifer S.	Newbenefits	Stafford
987654321	20	15.0	Wallace,Jennifer S.	Reorganization	Houston
888665555	20	null	Borg,James E.	Reorganization	Houston

Similar comments apply to the EMP_PROJ relation which augments the WORKS_ON relation with additional attributes from EMPLOYEE and PROJECT.

- Storing natural joins of base relations leads to an additional problems referred to as Update anomalies.

- Another serious problem with EMP_DEPT and EMP_PROJ is **Update anomalies**.

- **3 types of Update anomalies:**

1. Insertion Anomalies -> 2 types:

- a) Consistency problem
- b) Null value problem

2. Deletion Anomalies

3. Modification Anomalies

INSERTION ANOMALIES

- Insertion Anomalies can be differentiated into 2 types:
 1. When adding a new employee to the EMP_DEPT table, we must also enter department details. If the department information is unavailable, we must insert NULL values, which can cause inconsistencies.
 2. A new department cannot be added to the EMP_DEPT table unless at least one employee is assigned to it, making it difficult to store department details independently.

- **Problem 1:** We cannot add an employee without also adding department details.
- **Problem 2:** We cannot add a department unless it has at least one employee.

This issue happens because **EMP_DEPT** combines employee and department details in one table, leading to dependency problems.

insertion anomalies : consistency problem

Example : To insert a new tuple for an employee who works in department number 5, we must enter the attribute values of department 5 correctly so that they are consistent with values for department 5 in other tuples in EMP_DEPT.

EMP_DEPT						
ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER	DNAME	DMGRSSN
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

Key Issue:

- Every time we add a new employee, we must **manually ensure** that department details match the existing ones.
- This increases the risk of **inconsistencies** and **errors** in the database.

This problem happens because **EMP_DEPT** combines employee and department details in one table instead of keeping them separate.

consistency problem

In the design of the Figure 14.1 , we do not have to worry about this consistency problem because we enter only the department number in the employee tuple;

EMPLOYEE

ENAME	SSN	BDATE	ADDRESS	DNUMBER
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4
Narayan, Ramesh K.	666884444	1962-09-15	975 Fire Oak, Humble, TX	5
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1

DEPARTMENT

DNAME	DNUMBER	DMGRSSN
Research	5	333445555
Administration	4	987654321
Headquarters	1	888665555

- In this design, the **EMPLOYEE** table stores only the **department number (DNUMBER)** instead of repeating full department details.
- The **DEPARTMENT** table separately stores department information like **DNAME (Department Name)** and **DMGRSSN (Manager's SSN)**.
- This separation ensures that department details are stored **only once**, reducing errors and maintaining consistency.
- When adding a new employee, we only need to provide the department number, avoiding the risk of inconsistent department information.

This approach **prevents data duplication** and **eliminates consistency issues** that arise when the same information is stored multiple times.

null values problem

It is difficult to insert a new department that has no employees as yet in the EMP_DEPT relation. The only way to do this is to place null values in the attributes for employee. This causes a problem because SSN is the primary key of EMP_DEPT

EMP_DEPT

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER	DNAME	DMGRSSN
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

- In the **EMP_DEPT** table, every department is linked to at least one employee.
- If we want to add a **new department** that has no employees yet, we must enter **NULL values** for the employee details.
- However, this causes a problem because **SSN (Social Security Number)** is the **primary key**, and primary keys **cannot be NULL**.
- This means we **cannot store a department** in the table unless at least one employee is assigned to it.

null values problem

This problem does not occur in the design of Figure 14.1, because a department is entered in the DEPARTMENT relation whether or not any employees work for it, and whenever an employee is assigned to that department, a corresponding tuple is inserted in EMPLOYEE.

EMPLOYEE

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4
Narayan, Remesh K.	666884444	1962-09-15	975 Fire Oak, Humble, TX	5
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1

DEPARTMENT

DNAME	<u>DNUMBER</u>	DMGRSSN
Research	5	333445555
Administration	4	987654321
Headquarters	1	888665555

- **Null values problem** does not happen in this design because departments are stored separately in the **DEPARTMENT** table, even if they have no employees.
- Whenever a new employee joins a department, their details are added to the **EMPLOYEE** table, linking them to an existing department. This allows departments to be stored in the database even if they have no employees yet.

deletion anomalies

If we delete from EMP_DEPT an employee tuple that happens to represent the last employee working for a particular department, the information concerning that department is lost from the database. *This problem does not occur in the database of Figure 14.1 because DEPARTMENT tuples are stored separately.*

EMP_DEPT

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER	DNAME	DMGRSSN
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

modification anomalies

In `EMP_DEPT`, if we change the value of one of the attributes of a particular department-say, **the manager of department 5** -we must update the tuples of all employees who work in that department; otherwise, the database will become inconsistent.

EMP_DEPT

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER	DNAME	DMGRSSN
Smith,John B.	123456789	1965-01-09	731 Fondren,Houston,TX	5	Research	333445555
Wong,Franklin T.	333445555	1955-12-08	638 Voss,Houston,TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring,TX	4	Administration	987654321
Wallace,Jennifer S.	987654321	1941-06-20	291 Berry,Bellaire,TX	4	Administration	987654321
Narayan,Ramesh K.	666884444	1962-09-15	975 FireOak,Humble,TX	5	Research	333445555
English,Joyce A.	453453453	1972-07-31	5631 Rice,Houston,TX	5	Research	333445555
Jabbar,Ahmad V.	987987987	1969-03-29	980 Dallas,Houston,TX	4	Administration	987654321
Borg,James E.	888665555	1937-11-10	450 Stone,Houston,TX	1	Headquarters	888665555

GUIDELINE 2:

Design the base relation schemas so that no insertion, deletion, or modification anomalies are present in the relations. If any anomalies are present, note them clearly and make sure that the programs that update the database will operate correctly..

3. Null Values in Tuples

- When we group many attributes, if many of the attributes do not apply to all tuples in the relation, then we end up with many nulls in those tuples.
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- Moreover, nulls can have multiple interpretations, such as:
 1. The attribute does not apply to this tuple.
 2. The attribute value for this tuple is unknown.
 3. The value is known but absent; that is, it has not been recorded yet.

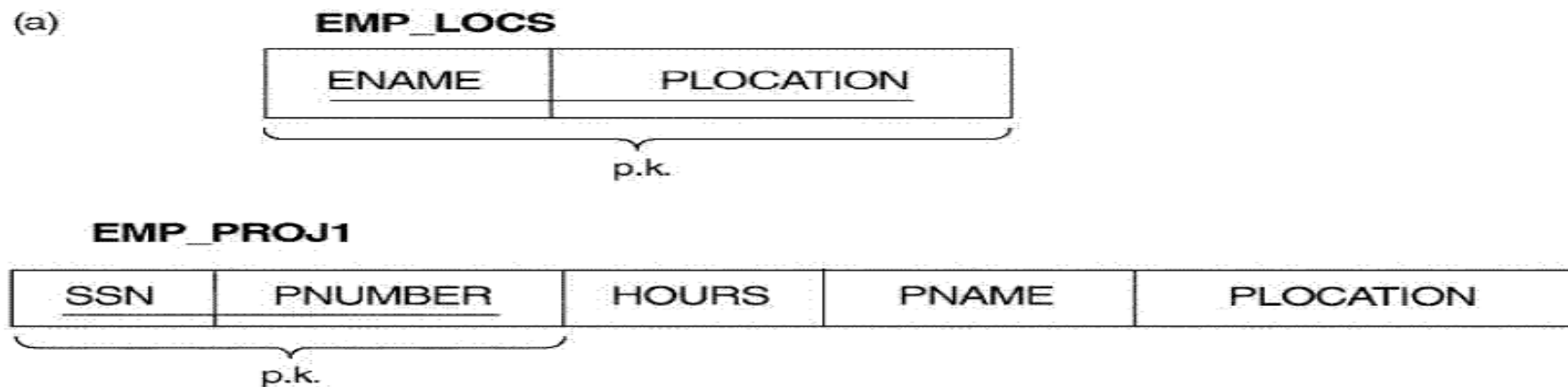
Guideline3 :

As far as possible, avoid placing attributes in a base relation whose values may frequently be null. If nulls are unavoidable, make sure that they apply in exceptional cases only and do not apply to a majority of tuples in the relation

4. Generation of Spurious Tuples

- A spurious tuple is basically, a record in a database that gets created when two tables are joined badly.

Let us split EMP_PROJ into 2 relation **EMP_PROJ1** and **EMP_LOCS** as shown below.



- A tuple in EMP_LOCS means that the employee whose name is ENAME works on *some project whose location is PLOCATION*
- A tuple in EMP_PROJ1 means that the employee whose social security number is SSN works HOURS per week on the project whose name, number, and location are PNAME, PNUMBER, and PLOCATION

Now produce tuples for **EMP_LOCS** by applying **project** operation to EMP_PROJ relation state , So we get tuples as shown below

EMP_LOCS	
ENAME	PLOCATION
Smith, John B.	Bellaire
Smith, John B.	Sugarland
Narayan, Ramesh K.	Houston
English, Joyce A.	Bellaire
English, Joyce A.	Sugarland
Wong, Franklin T.	Sugarland
Wong, Franklin T.	Houston
Wong, Franklin T.	Stafford

Zelaya, Alicia J.	Stafford
Jabbar, Ahmad V.	Stafford
Wallace, Jennifer S.	Stafford
Wallace, Jennifer S.	Houston
Borg, James E.	Houston

Using the **project operation** removes other details and keeps only the necessary columns.

Tuples for **EMP_LOCS** obtained by applying Project operation to EMP_PROJ

Now produce tuples for **EMP_PROJ1** by applying **project** operation to EMP_PROJ relation state , So we get tuples as shown below

EMP_PROJ1

SSN	PNUMBER	HOURS	PNAME	PLOCATION
123456789	1	32.5	Product X	Bellaire
123456789	2	7.5	Product Y	Sugarland
666884444	3	40.0	Product Z	Houston
453453453	1	20.0	Product X	Bellaire
453453453	2	20.0	Product Y	Sugarland
333445555	2	10.0	Product Y	Sugarland
333445555	3	10.0	Product Z	Houston
333445555	10	10.0	Computerization	Stafford
333445555	20	10.0	Reorganization	Houston
999887777	30	30.0	Newbenefits	Stafford
999887777	10	10.0	Computerization	Stafford
987987987	10	35.0	Computerization	Stafford
987987987	30	5.0	Newbenefits	Stafford
987654321	30	20.0	Newbenefits	Stafford
987654321	20	15.0	Reorganization	Houston
888665555	20	null	Reorganization	Houston

Tuples for **EMP_PROJ1** obtained by applying Project operation to EMP_PROJ

Now apply natural join operation (*) on **EMP_PROJ1** and **EMP_LOCS** to get the original relation **EMP_PROJ**. We consider only the tuples above dotted lines. So we get following relation

EMP_PROJ1 * EMP_LOCS					
SSN	PNUMBER	HOURS	PNAME	PLOCATION	
123456789	1	32.5	ProductX	Bellaire	Smith,John B.
* 123456789	1	32.5	ProductX	Bellaire	English,Joyce A.
123456789	2	7.5	ProductY	Sugarland	Smith,John B.
* 123456789	2	7.5	ProductY	Sugarland	English,Joyce A.
* 123456789	2	7.5	ProductY	Sugarland	Wong,Franklin T.
666884444	3	40.0	ProductZ	Houston	Narayan,Ramesh K.
* 666884444	3	40.0	ProductZ	Houston	Wong,Franklin T.
* 453453453	1	20.0	ProductX	Bellaire	Smith,John B.
453453453	1	20.0	ProductX	Bellaire	English,Joyce A.
* 453453453	2	20.0	ProductY	Sugarland	Smith,John B.
453453453	2	20.0	ProductY	Sugarland	English,Joyce A.
* 453453453	2	20.0	ProductY	Sugarland	Wong,Franklin T.
* 333445555	2	10.0	ProductY	Sugarland	Smith,John B.
* 333445555	2	10.0	ProductY	Sugarland	English,Joyce A.
333445555	2	10.0	ProductY	Sugarland	Wong,Franklin T.
* 333445555	3	10.0	ProductZ	Houston	Narayan,Ramesh K.
333445555	3	10.0	ProductZ	Houston	Wong,Franklin T.
333445555	10	10.0	Computerization	Stafford	Wong,Franklin T.
* 333445555	20	10.0	Reorganization	Houston	Narayan,Ramesh K.
333445555	20	10.0	Reorganization	Houston	Wong,Franklin T.

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:					
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Figure 14.06

Result of applying the NATURAL JOIN operation to the tuples above dotted lines in EMP_PROJ1 and EMP_LOCS, with generated spurious tuples marked by an asterisk.

Please note the following

- Additional tuples that were not in EMP_PROJ are called spurious tuples because they represent **spurious or *wrong information that is not valid.*** The *spurious* tuples are marked by asterisks (*).
- Decomposing EMP_PROJ into EMP_LOCS and EMP_PROJ1 is undesirable because, when we JOIN them back using NATURAL JOIN, we do not get the correct original information. This is because PLOCATION is neither a primary key nor a foreign key in either EMP_LOCS or EMP_PROJ1

GUIDELINE 4.

- Design relation schemas so that they can be joined with equality conditions on attributes that are either primary keys or foreign keys in a way that guarantees that no spurious tuples are generated.
- Avoid relations that contain matching attributes that are not (foreign key, primary key) combinations, because joining on such attributes may produce spurious tuples.

Summary of Design Guidelines

1. Semantics should be unambiguous and well defined .
2. Anomalies that imply additional work to be done during insertion into and modification of a relation, and that may cause accidental loss of information during a deletion from a relation.
3. Waste of storage space due to nulls and difficulty of performing aggregation operations and joins due to null values.
4. Generation of invalid and spurious data during joins on improperly related base relations